

Théo Gueuret

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Skills

- **Deep Learning / Vision:** PyTorch, PyTorch Geometric, PyTorch Lightning, TensorFlow, OpenCV, dlib
- **Computing / Data:** NumPy, pandas, scikit-learn, Spark
- **MLOps / Infra:** Git, Grid'5000, Docker, Slurm / OAR, Tensorboard
- **Programming languages:** Python (advanced), R / Scala / SQL (intermediate), C / Java (basics)
- **Writing & productivity:** LaTeX, Typst, AI-assisted development (Claude Code, Codex, GitHub Copilot)
- **Languages:** French (native), English (TOEIC 985/990), German (beginner), Japanese (beginner)

Education

- Université de Lille - CRISTAL, FOX team** Expected Sept. 2026
PhD in Computer Science - Graph deep learning for computer vision Lille, France
- Publications and talks at national and international conferences (EGC 2026, CBMI 2024)
 - **Best methodological paper nominee** at EGC 2026; invited extension to the Elsevier DKE journal (**Q1**)
- ESIEE Paris - Université Gustave Eiffel** Sept. 2018 — Aug. 2023
MSc in Engineering - Artificial Intelligence and Data Science - ranked 2nd/41 (final year) Champs-sur-Marne, France
- Exchange semester at Chalmers University of Technology, Sweden (2022)

Experience

- PhD Candidate** Oct. 2023 — Present
Université de Lille Lille, France
- Designed a family of **spatio-temporal GNN architectures** (relational convolution, multi-branch landmark/muscle fusion, graph topology learning) for dynamic facial expression recognition from facial landmark graphs
 - **Systematic ablation study** of the pipeline on 6 datasets (DFEW, FERV39K, AFEW, MAFW, Oulu-CASIA, CK+) with subject-independent cross-validation: critical components — inter-ocular normalization (+18.4 accuracy points), a 53-landmark subset outperforming MediaPipe's 468
 - **Lightweight and interpretable** approach (GradCAM / node-importance analysis linked to FACS action units), an efficient alternative to large pre-trained full-image models
- Deep Learning R&D Engineer - Graduation internship** Feb. 2023 — Aug. 2023
Capgemini Engineering Meudon, France
- Designed a **multimodal biometric authentication** system using siamese networks, CNN autoencoders and LSTMs
 - Published in the ELCVIA journal
- Research Intern - Computer Vision** May 2022 — Aug. 2022
University of Nevada (UNLV) Las Vegas, NV, USA
- Developed a **semi-supervised CNN** architecture for audio scene classification with limited data (TUT 2018), improving generalization over supervised CNN baselines ([github](#))
 - **Presented the results** as a poster at IEEE SEC 2022 (DOI: 10.1109/SEC54971.2022.00044)

Publications

Full list of publications available on [Google Scholar](#).

- **Journal** - "Dynamic Facial Expression Recognition via Spatio-temporal Graph Networks." *Advances in Knowledge Discovery and Management*, Elsevier DKE (**Q1**, submitted 2026) - invited following the EGC 2026 nomination
- **Conference** - "Reconnaissance d'expressions faciales dynamiques par réseaux de graphes spatio-temporels." EGC 2026, Anglet, France - **Best methodological paper nominee**
- **Conference** - "A Survey on Graph Deep Representation Learning for Facial Expression Recognition." CBMI 2024. DOI: 10.1109/CBMI62980.2024.10858877

Projects & Interests

- **Abandoned luggage project** — Object detection (YOLO) and IoU-association multi-target tracking to identify abandoned luggage, in real time or offline ([github](#))
- **Projet Sandy** — Software and robotic prototype for detecting, identifying and quantifying micro-plastics in sand. Winner of the ESIEE Paris Projects Day 2021, **Health-Environment-Disability Prize** ([github](#))
- **Chess** (competitive, ~ 1700 Elo, *Lille Métropole Chess*), **Badminton** (competitive, *LUC Badminton*), **Cycling**, **Running**
- **Chessminton** — Organizer of the 1st Lille Open (2026), a hybrid chess/badminton format (LUC & LMC)